

Unit 3: Introduction to Geology & Geologic Time

Reading: pages 39–50 · 20 points · 15 minutes · Closed book

Name _____

Date _____

Score / 20

Answer every question. For multiple choice, circle the single best answer. Write short answers in complete sentences.

Part A · Multiple Choice

6 questions × 2 points = 12 points

- Which principle states that in an otherwise undisturbed sequence of sedimentary layers, the oldest layers are on the bottom?
 - Principle of Superposition
 - Principle of Lateral Continuity
 - Principle of Fossil Succession
 - Principle of Original Horizontality
- James Hutton's principle of uniformitarianism is usually summarized as
 - "the present is the key to the past"
 - "the rocks never lie"
 - "all strata were once horizontal"
 - "life succeeds life"
- Geological time is divided into four eons. Which is the most recent, and takes its name from words meaning "visible life"?
 - Hadean
 - Archean
 - Proterozoic
 - Phanerozoic
- The three eras of the Phanerozoic, listed from oldest to youngest, are
 - Cenozoic, Mesozoic, Paleozoic
 - Paleozoic, Mesozoic, Cenozoic
 - Mesozoic, Paleozoic, Cenozoic
 - Paleozoic, Cenozoic, Mesozoic
- The half-life of carbon-14 is 5,730 years. Since about ten half-lives is the practical limit of the method, radiocarbon dating is useful back to roughly
 - 5,730 years
 - 11,460 years
 - 57,300 years
 - 573,000 years

6. After exactly two half-lives have passed, a mineral sample contains
- a) 50% parent and 50% daughter
 - b) 25% parent and 75% daughter
 - c) 75% parent and 25% daughter
 - d) 0% parent and 100% daughter

Part B · True or False

2 questions × 1 point = 2 points

7. Relative dating tells a geologist the specific numerical age of a rock in years.
TRUE / FALSE
8. William Smith is credited with formulating the principle of faunal succession.
TRUE / FALSE

Part C · Short Answer

2 questions × 3 points = 6 points

9. Name the instrument used to separate isotopes in radio-isotopic dating, and explain how an age is calculated from a rock sample.
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10. The text calls geology “arguably the most integrated of all of the sciences.” Explain why, and name the extra dimension geology has that most other sciences do not.
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